CDT204 – Computer Architecture

> Date: Aug. 22nd 2024
>
> Time: 8:30 – 12:30
>
> Help: Small calculator is allowed in the exam

The exam has 37 points. The grades will be awarded as follows:

- 3 : 19 Points
- 4 : 28 Points
- 5 : 34 Points

Important Notes:

- Give as full an answer as possible to obtain full marks. All calculations, approximations, assumptions and justifications must be reported for full credit unless stated otherwise. Please use figures and examples to clarify.
- If you do not understand a question clearly, you are allowed to call the teacher and ask.
- Write the question and part number on each page clearly.
- Answer each question on a separate page.
- In case you might have forgotten:

  1GB = 2^{10}MB = 2^{20}KB = 2^{30}B

  1 sec = 10^3 ms = 10^6 μs = 10^9 ns = 10^{12} ps

  $$S = \frac{T_0}{\frac{T_A}{F} + (T_0 - T_A)}$$

Live long and prosper

Task 1 (4p)

Signed integer numbers are stored in the computer memory using "Two's Complement" method.

a) Explain this method with examples. (2p)
b) Why is "two's complement" the most widely used method for storing signed integers? (2p)

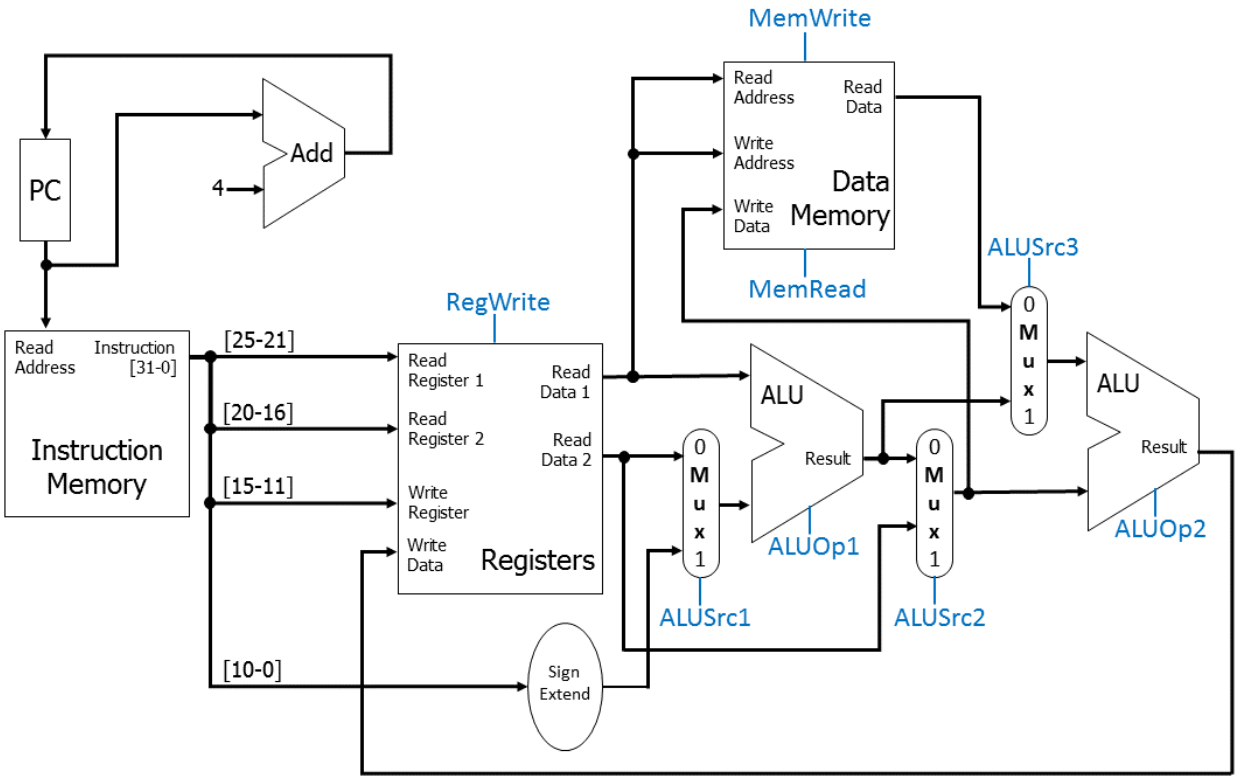
Task 2 (8p)

Consider the following sequence of MIPS instructions:

```
SW   R16, -100(R6)
LW   R4, 8(R16)
ADD  R5, R4, R4
```

a) Indicate dependencies and their type. (2p)
b) Assume there is no forwarding in this pipelined processor. Indicate hazards and add NOP instructions to eliminate them. (2p)
c) Assume there is full forwarding. Indicate hazards and add NOP instructions to eliminate them. (2p)
d) What is the total execution time of this instruction sequence without forwarding and with full forwarding? What is the speedup achieved by adding full forwarding to a pipeline that had no forwarding? (Assume that the latency for all the stages is equal and is 250ps for non-forwarding system and 300ps with full forwarding). (2p)

Task 3 (8p)



The datapath above supports the following instructions:

- **`lw_add rd, (rs), rt`** (fused load and add): Register `rd` will be set to the summation of memory value addressed by register `rs` and the value stored in register `rt`. `R[rd] = Mem[R[rs]]+R[rt]`
- **`addi_st (rs), rs, imm`** (fused immediate add and store): The memory addressed by register `rs` will be set to summation of register `rs` and constant value `imm`. `Mem[R[rs]]=R[rs]+imm`
- **`sll_add rd, rs, rt, imm`** (fused immediate left shift and sum): Value of register `rs` is shifted left by constant value `imm` then added to register `rt` and stored in register `rd`. `R[rd]=(R[rs]<<imm)+R[rt]`

All instructions use the same format (shown below). Not all instructions use all the fields below.

op (bits 31-26)	**rs** (bits 25-21)	**rt** (bits 20-16)	**rd** (bits 15-11)	**imm** (bits 10-0)

For each of the above instructions, specify how the control signals should be set for correct operation. Use `X` for don't care. `ALUOp` can be `ADD`, `SUB`, `SLL`, `PASS_A`, or `PASS_B` (i.e. `PASS_A` means pass the top operand through without changing it).

You can find an empty table for this task at the last page of the exam sheet, which you can use to write your answer in. Detach it from the exam sheet and put it in your answers folder.

Task 4 (6p)

Consider the following C code. Assume that each element of the two-dimensional arrays A and B is a 64-bit integer. Assume that cache block size is 16 bytes.

```
for(i=0; i<1024; i++)
  for(j=0; j<1024; j+=2)
        A[i][j] = B[i][0] + A[j][i];
```

A. Explain temporal and spatial locality. (2p)
B. Which variable references exhibit temporal locality? (2p)
C. Which variable references exhibit spatial locality? (2p)

Task 5 (8p)

You have been asked to improve the performance of an application developed by another programmer. In this application, one of the most executed parts of the code is as follows.

```
bool shot_by_laser[entries];

for(int i=0; i< entries; i++){

  if(objects.isCharacter[i]==true && objects.isAlive[i]==true)
        shot_by_laser[i] = true;
  else
        shot_by_laser[i] = false;

}
```

Assume that:

- All the variables are four-byte long.
- The number of entries is 32768 (2^{15}).
- The processor has a direct-mapped cache of 8MB with a cache block size of 64 bytes.
- A cache hit takes one cycle and a cache miss takes 100 cycles for both load and store.
- The starting address of `objects.isCharacter`, `objects.isAlive`, and `shot_by_laser` are 0x5000000, 0x06000000, and 0x07000000 respectively.
- The execution time of non-memory instructions is zero.

A. How many cycles are required to run this code? (5p)
B. What suggestions you might have to improve this code? (3p) Explain.

Task 6 (3p)

Below some benchmarks testing two different CPUs are presented. The Cinebench Multicore tests a CPU's multicore performance where all the cores on the CPU are asked to perform heavy calculations. Cinebench Single core is the same as the first one but instead it only uses one of the cores on the CPU. The last benchmark (Factorio) is a game. (Benchmarks are taken from techspot.com)

Processor	AMD Ryzen 7 7800X3D	AMD Ryzen 7 7700X
Cinebench Multicore	Score: 18395	Score: 19910
Cinebech Single core	Score: 1816	Score: 1969
Factorio	FPS: 429	FPS: 242

The specs for these two CPUs are presented in the following table.

Processor	AMD Ryzen 7 7800X3D	AMD Ryzen 7 7700X
No. of Cores/Threads	8 Cores / 16 Threads	8 Cores / 16 Threads
L1 Cache Size	64 KB (per core)	64 KB (per core)
L2 Cache Size	1 MB (per core)	1 MB (per core)
L3 Cache Size	96 MB	32 MB
Base Clock	4.2 GHz	4.5 GHz
Boost Clock	5.0 Ghz	5.4 Ghz

A. What can you deduct from Cinebech Multicore (left) benchmark, comparing the two CPUs? (1p)
B. What can you deduct from Cinebech Single core (middle) benchmark, comparing the two CPUs? (1p)
C. In the 3rd benchmark (Factorio, right) the 7800X3D renders almost double as the frames as the 7700X which was the faster one so far. Why? (1p)

Answer Sheet for Task 3

Inst.	ALUSrc1	ALUSrc2	ALUSrc3	ALUOp1	ALUOp2	MemRead	MemWrite	RegWrite
lw_add								
addi_st								
sll_add								